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ECE 4644 / 5610 Satellite Communications

Homework Assignment #3 SOLUTIONS

Note: For all problems take the radius of Earth, R_E , to be 6378.137 km.

Problem #1

Consider that satellites in the Starlink mega-constellation orbit the earth at 550 km altitude.

a. Determine the length of the arc along Earth's surface that can be viewed by a Starlink satellite to the limit of the horizon. (The solution of HW#1, Prob #1 might be helpful)

b. Suppose the satellites are orbiting in distinct shells that are separated in longitude and that the satellites are matched across the shells so that they cross the equator at the same time. What minimum number of satellites need to cross the equator at the same time to provide coverage around the entire equator at that moment? What is the minimum number of orbital shells that is required?

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c. The actual number of Starlink orbital planes is 24. Comment.

SpaceX has announced that thousands of satellites will be lowered in altitude to 480 km over the course of 2026.

d. With the satellites at lower altitude, each will cover a smaller area. Estimate the factor (ratio) by which the coverage area will be reduced. What is the implication for the number of satellites needed to cover the same total area as before?

e. The inclination angle of the Starlink orbit is 53°. What are the highest and lowest latitudes where the satellites can be seen passing directly overhead? (This is the most favorable situation for establishing links to ground stations.) Why do you think this inclination angle was chosen?

Problem #2

a. In your own words and with the help of a sketch depicting the motion of the Earth around the Sun, explain the difference between a *mean solar day* and a *sidereal* day. Assuming that the Earth's orbit is a circle and a complete revolution around the sun takes 365.24 days, show that a solar day is longer by about 4 min.

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b. The planet Zorg was just discovered revolving around the Sun in Earth's orbit but on the exact opposite side – it wasn't detected before the space age because the Sun blocks our view of it from the Earth. Zorg has the same length year as Earth (obviously) and the same length sidereal day but it rotates about its axis in the opposite sense to Earth, so that the Sun rises in the west and sets in the east. What is the length of a mean solar day on Zorg? (This is called retrograde rotation. Two planets in the solar system, Venus and Uranus, exhibit retrograde rotation.)

Problem #3

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In a few paragraphs, explain why satellites do not follow pure Keplerian orbits. Cite the major sources of perturbation to satellites in GEO and LEO orbits, respectively, and describe the sorts of maneuvers that are required to deal with the consequences, again separately for satellites in GEO and LEO orbits. (Avoid excessive detail.)

Problem #4

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Describe and sketch three methods for raising a payload into GEO. Use the words *direct*, *GTO*, *AKM*, *slow*) and discuss their relative advantages

Problem #5

We will review the problem of communication with the Apollo missions in lunar orbit. Specifically, we are concerned with the length of the radio blackout that occurred when the spacecraft in lunar orbit went behind the moon (as seen from Earth).

a. Taking the mass of the moon to be $7.3477\ 10^{22}$ kg, find the value of Kepler's constant for the moon, μ_L . It might be helpful to know that the ratio of masses is $M_E / M_L = 81.30$.

b. The orbit of the Apollo s/c was approximately circular at an altitude of 110 km. Taking the radius of the moon to be $R_L = 1{,}737.4$ km, determine the semi-major axis of the orbit and its period.

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c. Assume for simplicity that the moon and s/c are confined to the equatorial plane of Earth so that this geometry is two-dimensional. Sketch the Earth, moon, and satellite orbit looking down from far above Earth's north pole. Take the distance between the centers of Earth and moon to be 385,000 km. (Drawing strictly to scale is not necessary but it will be helpful to show that Earth-moon distance is large compared to the sizes of the bodies.)

d. Communication with the Apollo mission was by ground stations. Assuming that ground stations are available continuously around the perimeter of Earth, illustrate the blackout region on the far side of the moon by drawing lines from endpoints of the Earth 'baseline' to the horizon points on the moon (two lines). Extend these lines to show the space behind the moon where communication will not be possible. (This will look similar to an eclipse plot with radio blackout happening when the moon totally eclipses Earth.)

+ 2

e. Estimate the amount of time the s/c would be in radio blackout per orbit. (Bonus for exact answer. This will involve some geometrical analysis.)

$\Sigma = 35\ (+2)$

ECE 5610: Satellite Communications
Homework #3

Problem #1. Starlink satellites at altitude $h = 550$ km above Earth. Take $R_E = 6378.137$ km. Let $r = R_E + h$ be the satellite orbital radius.

(a) Arc length on Earth visible to the horizon

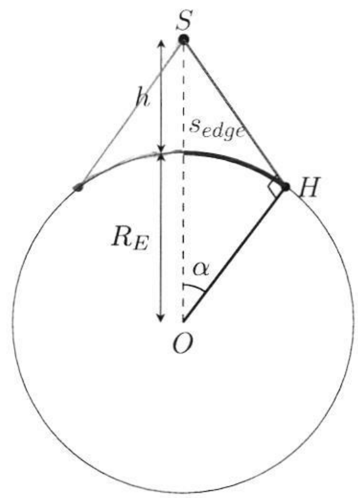


Figure 1: Geometry of the horizon-limited visibility for a satellite at altitude h.

At the horizon, the line-of-sight from the satellite is tangent to Earth. In the right triangle formed by Earth center O, satellite S, and tangent point H,

$$OS = r, \quad OH = R_E, \quad \angle H = 90^\circ.$$

Let α be the Earth-central angle from the subsatellite point to the horizon point. Then

$$\cos\alpha = \frac{OH}{OS} = \frac{R_E}{R_E + h} \quad \Rightarrow \quad \alpha = \cos^{-1}\Big(\frac{R_E}{R_E + h}\Big).$$

For $h = 550$ km, $r = 6928.137$ km, so

$$\alpha = \cos^{-1}\Big(\frac{6378.137}{6928.137}\Big) = 0.40114 \text{ rad} = 22.984^\circ.$$

Ground arc length (subsatellite point to horizon) is

$$s_{\text{edge}} = R_E\alpha = 6378.137(0.40114) = 2558.6 \text{ km}.$$

Thus the full horizon-to-horizon arc (through the subsatellite point) is

$$s_{\text{visible}} = 2R_E\alpha = 5117.1 \text{ km}.$$

✓

(b) Minimum simultaneous equator-crossers for full equator coverage

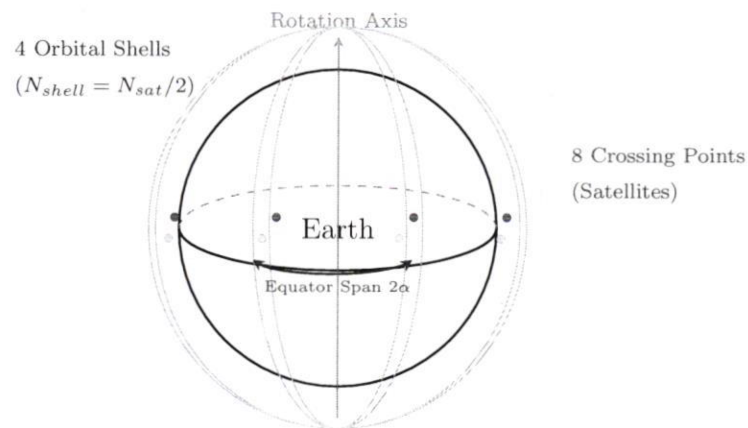


Figure 2: Visualizing the relationship between orbital shells and equatorial coverage. One shell provides two coverage points (nodes) on the equator.

The circumference of the equator is $C_E = 2\pi R_E$. At the instant a satellite crosses the equator, its horizon-limited coverage along the equator is approximately the same horizon-to-horizon arc length $L_s = 2R_E\alpha$. Therefore the minimum number of satellites that must cross the equator *simultaneously* to cover 360° is ✓

$$N_{\min} = \left\lceil \frac{C_E}{L_s} \right\rceil \approx \left\lceil \frac{360°}{2\alpha°} \right\rceil = \left\lceil \frac{360}{2(22.984)} \right\rceil \approx 8.$$

To convert satellites to *orbital shells/planes*, note that a single orbital plane intersects the equator at *two* nodes (ascending and descending) separated by 180°. With satellites phased so that both node crossings occur at that instant, each plane can provide two of the equator-crossing satellites. Thus,

$$N_{\text{shell,min}} = \left\lceil \frac{N_{\min}}{2} \right\rceil = \left\lceil \frac{8}{2} \right\rceil = 4.$$

✓

(c) Comment on the actual value: 24 orbital planes

The value in (b) is a *geometric lower bound* based on horizon-limited links. In practice, many more planes are used because users generally need satellites at *higher elevation angles* (not near the horizon), especially in urban/obstructed environments. In addition, a larger number of planes improves **capacity** (more simultaneous links) and **reliability/performance** (redundancy against outages, smoother handovers, and less sensitivity to individual satellite/plane gaps). Hence an operational design such as 24 planes can be much larger than the minimum required by pure geometry.

with our design we have assumed operation down to el ∠ = 0°, with 6 x the orbital planes operation will likely be assured at high (>70°) elevation angles

(d) Coverage-area reduction when lowering altitude to $h = 480$ km

For a horizon-limited footprint on a spherical Earth, the visible region is a spherical cap of angular radius α, whose area is

$$A = 2\pi R_E^2(1 - \cos\alpha), \quad \text{with} \quad \cos\alpha = \frac{R_E}{R_E + h}.$$

Compute α at $h_1 = 550$ km and $h_2 = 480$ km:

$$\alpha_1 = \cos^{-1}\Big(\frac{R_E}{R_E + 550}\Big) = 22.984°, \qquad \alpha_2 = \cos^{-1}\Big(\frac{R_E}{R_E + 480}\Big) = 21.564°.$$

✓

Therefore the area ratio is

$$\frac{A_2}{A_1} = \frac{1 - \cos\alpha_2}{1 - \cos\alpha_1} = \frac{1 - \frac{R_E}{R_E+480}}{1 - \frac{R_E}{R_E+550}} = 0.8816.$$

So the coverage area becomes $\approx 0.882 A_1$, i.e., about a 11.8% reduction. Implication: to cover the *same* total area (all else equal), the required satellite count scales approximately as $1/A$, so

$$\frac{N_2}{N_1} \approx \frac{A_1}{A_2} = \frac{1}{0.8816} \approx 1.134,$$

✓

meaning about $\boxed{13\%}$ more satellites for the same total coverage area.

(e) Highest/lowest latitudes for passing directly overhead; why 53°? A satellite passes

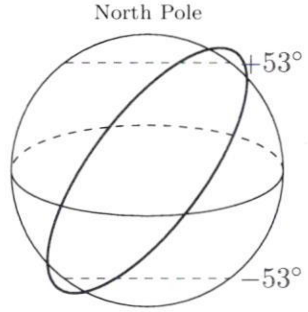


Figure 3: Orbital inclination $i = 53°$ and the resulting latitude limits for zenith passes.

directly overhead at a location only if that location lies on the ground track. For a circular orbit of inclination i, the ground track reaches latitudes from $-i$ to $+i$. Thus, for $i = 53°$,

$$\boxed{\text{Highest overhead latitude } = +53°, \qquad \text{Lowest } = -53°.}$$

✓

A 53° inclination concentrates coverage on mid-latitudes where a large fraction of the world's population lives, while requiring fewer satellites than a near-polar design for the same mid-latitude performance; it is a practical trade between geographic coverage and constellation size/capacity.

approximately 98-99% according to Gemini chatbot ✓

Problem #2

(a) Mean solar day vs. sidereal day

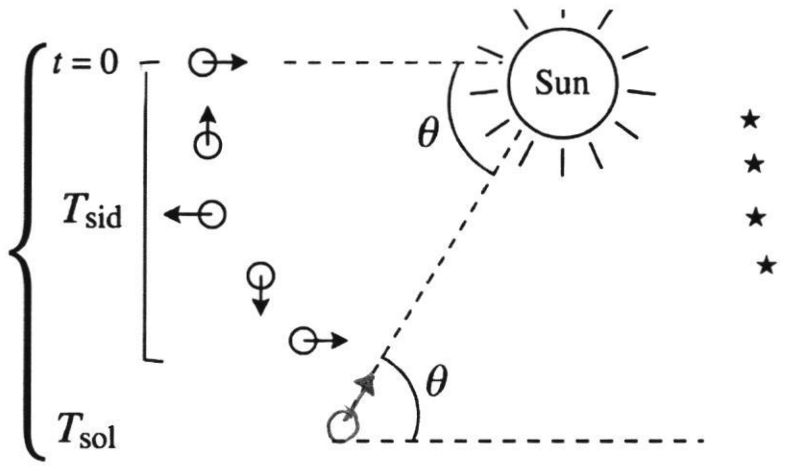


Figure 4: In one solar day the Earth must rotate $360° + \theta$ to bring the Sun back overhead, while in one sidereal day it rotates $360°$ relative to the stars.

(a) Show that $T_{\text{sid}} < T_{\text{sol}}$ by ≈ 4 minutes.

A *sidereal day* is the time for Earth to rotate $360°$ relative to distant stars. A *solar day* is noon-to-noon (Sun directly overhead again). Because Earth advances along its orbit while it rotates, the Sun direction shifts slightly, so Earth must rotate a little *more* than $360°$ (relative to the stars) to reach the next noon. ✓

Let θ be Earth's orbital advance (as seen from the Sun) during one *solar* day (see Fig. 4). Since Earth goes $360°$ around the Sun in 365.24 solar days,

$$\theta = \frac{360°}{365.24}.$$ ✓

Thus, in one solar day the Earth must rotate

$$360° + \theta$$

relative to the stars to bring the Sun back overhead, whereas in one sidereal day it rotates exactly $360°$. Because the rotation rate is the same physical rate,

$$\frac{360°}{T_{\text{sid}}} = \frac{360° + \theta}{T_{\text{sol}}} \quad \Longrightarrow \quad T_{\text{sid}} = T_{\text{sol}} \frac{360°}{360° + \theta}.$$

With $T_{\text{sol}} = 24$ h and $\theta = 360°/365.24$,

$$T_{\text{sid}} = 24\text{ h} \cdot \frac{360}{360 + 360/365.24} = 24\text{ h} \cdot \frac{365.24}{366.24} = 23.93447\text{ h} \approx 23\text{ h } 56\text{ min } 4\text{ s}.$$

Therefore,

$$T_{\text{sol}} - T_{\text{sid}} \approx 24\text{ h} - 23\text{ h } 56\text{ min } 4\text{ s} \approx 3\text{ min } 56\text{ s} \approx 4\text{ min}.$$ ✓

(b) Mean solar day on Zorg (retrograde rotation)

On Earth (prograde rotation), the planet must rotate *more* than 360° to complete a solar day because its orbital motion and rotational motion are in the same direction. On Zorg, however, the rotation is retrograde. As Zorg moves from Point A to Point B in its orbit, its "backward" rotation causes a fixed meridian to face the Sun *sooner* than it would complete a full 360° rotation relative to the stars.

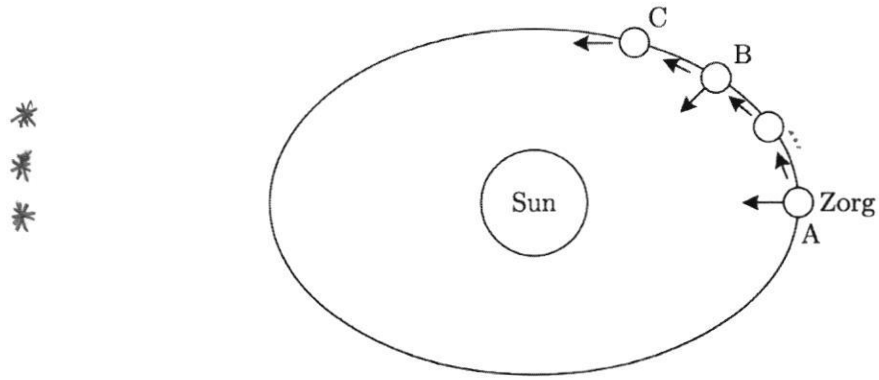


Figure 5: On Zorg, the retrograde rotation means the planet faces the Sun (Point B) before it completes a full sidereal rotation (Point C). Thus, $T_{\text{sol}} < T_{\text{sid}}$.

In part (a), we established that for a prograde planet like Earth, the planet must rotate $360° + \theta$ to complete one solar day, where $\theta \approx 1°$ is the orbital advance per day. ✓

Because Zorg has *retrograde* rotation, it rotates in the opposite direction of its orbital motion. As shown in the figure, this means the meridian faces the Sun **before** it completes a full 360° sidereal rotation. Specifically, it only needs to rotate:

$$360° - \theta$$

relative to the stars to bring the Sun back overhead. Since the rotation rate is the same physical rate as Earth's, we simply flip the sign of the orbital correction from part (a):

$$T_{\text{sid}} = T_{\text{sol,Z}} \cdot \frac{360°}{360° - \theta} \implies T_{\text{sol,Z}} = T_{\text{sid}} \cdot \frac{360° - \theta}{360°}$$

Using the values from part (a) where $\theta = 360°/365.24$:

$$T_{\text{sol,Z}} = 24 \text{ h} \cdot \frac{365.24}{367.24}$$ ✓

$$T_{\text{sol,Z}} \approx 23.86930 \text{ h} \approx 23 \text{ h } 52 \text{ min } 9 \text{ s}$$

Just as Earth's solar day is $\approx$ 4 minutes *longer* than its sidereal day (23h 56min + 4min), Zorg's solar day is $\approx$ 4 minutes *shorter* (23h 56min − 4min).

$$\boxed{\text{Mean solar day on Zorg } \approx 23 \text{ h } 52 \text{ min } 9 \text{ s}}$$

Problem #3.

Why satellites do not follow pure Keplerian orbits. Kepler's laws give a compact geometric description of ideal orbits, and Newton's law of gravitation provides the physical foundation. In our basic (two-body) model, we implicitly assume a *perfectly spherical, uniformly distributed* Earth and a satellite that feels the *same* central inverse-square gravitational force everywhere along its path. In practice, additional forces and nonuniform gravity perturb the orbit:

- **Other gravitating bodies:** the Moon and the Sun exert additional gravitational pulls (third-body effects) that perturb the orbit.
- **Non-spherical, non-uniform Earth:** Earth is oblate and its mass is not perfectly uniform, so the gravity field is not identical at every orbital position (e.g., J_2, tesseral/sectoral harmonics).
- **Non-gravitational forces:** atmospheric drag (especially in LEO), solar radiation pressure (SRP), and even weaker effects such as solar wind/plasma interactions at higher altitudes.

Because of these perturbations, the classical orbital elements $(a, e, i, \Omega, \omega, M)$ are not strictly constant; instead, the spacecraft follows an *osculating Keplerian orbit* whose elements slowly drift and undergo periodic variations.

Major perturbations and required maneuvers in GEO. In GEO, atmospheric drag is negligible. The dominant perturbations are

- **Third-body gravity (Moon and Sun):** gradually perturbs the orbital plane, leading to inclination drift.
- **Non-spherical Earth gravity field:** causes longitude (east–west) drift and slow changes in the orbit shape/orientation if not corrected.
- **Solar radiation pressure (SRP):** can also bias the eccentricity vector and produce periodic variations.

To maintain a GEO "station-keeping box," operators typically perform:

- **North–South station keeping** to control inclination (countering Moon/Sun effects).
- **East–West station keeping** to control longitude drift and manage small eccentricity-vector effects.

Major perturbations and required maneuvers in LEO. In LEO, the dominant perturbation is

- **Atmospheric drag:** reduces orbital energy, causing semi-major axis (altitude) decay and changing the period over time.

A second important effect is

- **Earth oblateness (J_2) and related gravity terms:** produces predictable precession of the node Ω and argument of perigee ω.

As a consequence, LEO satellites commonly require:

- **Reboost/drag makeup maneuvers** (or continuous low-thrust for very low LEO) to maintain altitude and mission lifetime.
- **Occasional orbit maintenance/phasing adjustments** to maintain ground-track timing or constellation spacing (as needed by the mission).

Problem #4. Describe and sketch three methods for raising a payload into GEO (*direct*, *GTO*, *AKM*, *slow*) and discuss relative advantages.

Three common methods

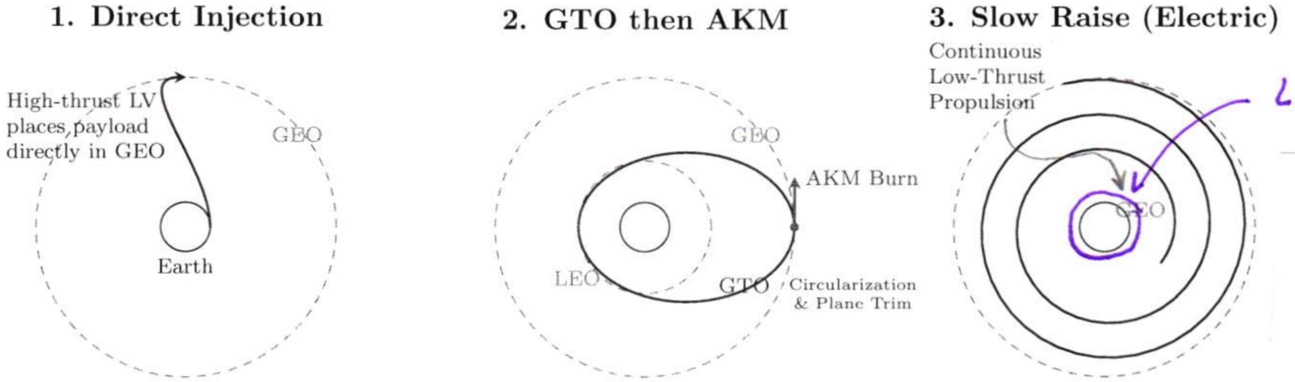


Figure 6: Comparison of GEO raising methods: (1) Direct injection requires high-thrust launchers; (2) GTO uses an AKM for efficient transfer; (3) Slow spiral utilizes high-efficiency electric propulsion.

(1) Direct insertion. A multi-stage launch vehicle provides essentially the full Δv and injects the spacecraft directly into (or very near) GEO. *Pros:* fastest approach and minimal orbit-raising required by the spacecraft (primarily station keeping afterward). *Cons:* requires a high-performance launcher; typically more demanding (and costly) in launch capability per delivered payload to GEO. ✓

(2) Staged transfer: LEO checkout → GTO → GEO ("GTO" + "AKM"). The launcher first inserts the spacecraft into a low circular orbit for checkout, then performs a transfer into an elliptical **GTO** with perigee near LEO and apogee near GEO altitude. Near apogee, the spacecraft fires an **apogee kick motor (AKM)** to circularize ? into GEO; this burn can also reduce inclination toward $i \approx 0°$ (geo-insertion). *Pros:* efficient and widely used; reduces the launch vehicle burden and offers operational flexibility. *Cons:* requires onboard propellant and a critical apogee burn; any needed inclination reduction increases the total Δv; where, Δv denotes the total velocity increment required from propulsion burns to transfer between orbits (units of m/s). ✓

checkout

GTO

Start again with LEO

(3) Slow orbit raising using electric propulsion. Rather than a single large apogee kick, the spacecraft uses continuous low thrust (e.g., ion/Hall/arcjet) to gradually raise the orbit and circularize at GEO, typically over weeks to months. *Pros:* high propellant efficiency, allowing more mass for payload or longer operational life. *Cons:* long transfer duration and increased operational constraints while thrusting.

explain

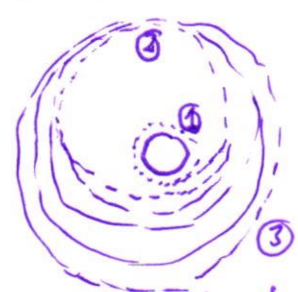
how long?

Note. Launching eastward and closer to the equator provides the largest rotational velocity boost and reduces the initial inclination for GEO missions, lowering the Δv required for geo-insertion.

explain

Slow Raise (Classic)

1) LEO orbit
2) GTO orbit
3) GEO orbit reached by continuous firing of ion thrusters to boost perigee and circularize the orbit



Problem #5

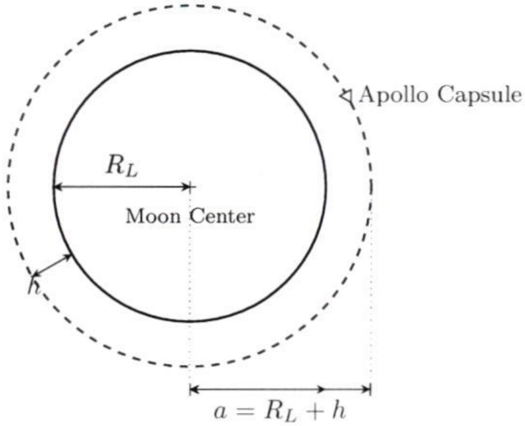
(a) Kepler's Constant for the Moon

Kepler's constant μ_L is calculated using the gravitational constant G and the Moon's mass M_L:

$$\mu_L = G \cdot M_L = (6.674 \times 10^{-11}\ \mathrm{m^3/kg \cdot s^2}) \times (7.3477 \times 10^{22}\ \mathrm{kg}) \approx 4.9028 \times 10^{12}\ \mathrm{m^3/s^2}$$

✓

(b) Orbital Parameters



The semi-major axis a for a circular orbit at altitude $h = 110$ km is:

$$a = R_L + h = 1737.4 + 110 = 1847.4\ \mathrm{km} = 1.8474 \times 10^6\ \mathrm{m}$$ < express in km

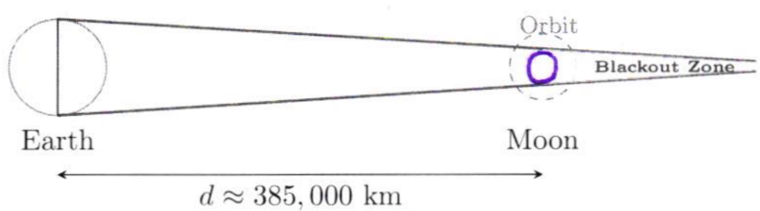
The orbital period T is given by Kepler's Third Law:

✓

$$T = 2\pi\sqrt{\frac{a^3}{\mu_L}} = 2\pi\sqrt{\frac{(1.8474 \times 10^6)^3}{4.9028 \times 10^{12}}} \approx 7130.6\ \mathrm{s} \approx 1.98\ \text{hours} \approx \textbf{1 hour, 58 minutes, and 51 seconds}$$

(c) & (d) Geometric Sketch of Radio Blackout

The diagram below illustrates the 2D coplanar geometry. The "Blackout Zone" is the conical region behind the Moon where the line of sight to the entire Earth is obstructed by the Moon's physical body.



✓

(e) Blackout Duration

Quick estimate: In the coplanar "worst-case" geometry, the capsule is behind the Moon for roughly half of each orbit, so

✓

$$t_{\text{blk}} \approx \frac{T}{2} \approx \frac{7130.6}{2} \approx 3565\ \mathrm{s} \approx 59\ \mathrm{min}.$$

(e) Bonus: Intuitive Geometric Analysis of Radio Blackout

The Concept of the Exit Angle (α)

If the Apollo capsule were traveling exactly on the lunar surface ($h = 0$), it would spend half its orbit (180°) in total darkness. However, because the capsule orbits at an altitude of 110 km, it can "see" over the curve of the Moon. This creates a "delay in eclipse," meaning the capsule stays in contact with Earth longer than a surface-level observer.

The **exit angle** α is the angular offset from the Moon's vertical pole to the point where the capsule finally crosses the shadow boundary. As shown in Figure 7, this angle is defined by the ratio of the shadow's starting height (the Moon's radius R_L) to the capsule's orbital radius (a):

$$\cos(\alpha) = \frac{R_L}{a} = \frac{1737.4 \text{ km}}{1847.4 \text{ km}} \approx 0.9405 \implies \alpha \approx 19.92^\circ$$

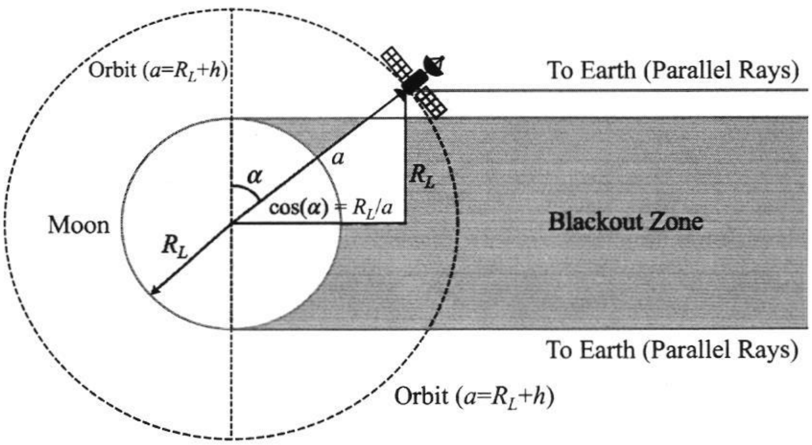


Figure 7: **Geometry of the Lunar Blackout Zone.** The shadow region (grey) is bounded by parallel rays from Earth tangent to the Moon. The capsule orbits at radius a. The exit angle α represents the "extra" arc the capsule travels in the light before entering or after leaving the shadow.

Refined Blackout Duration

Because the capsule extends visibility by α at entrance/exit, the total blackout arc is reduced from 180° to:

$$\Delta\theta_{\text{blk}} = 180^\circ - 2\alpha = 180^\circ - 2(19.92^\circ) \approx 140.16^\circ$$

The blackout duration is the time required for the capsule to sweep through this reduced angle. Using the orbital period $T \approx 118.84$ minutes calculated in part (b):

$$t_{\text{blk}} = T \cdot \frac{140.16^\circ}{360^\circ} = 118.84 \text{ min} \cdot 0.3893 \approx \mathbf{46.27} \text{ minutes}$$

Conclusion

This geometric correction accounts for the spacecraft altitude $h \neq 0$. By being 110 km above the surface, the astronauts gain approximately 13 minutes of communication time per orbit compared to the simple 59.4-minute estimate. This coplanar model represents the "worst-case" maximum blackout; any orbital inclination would further reduce this duration.

$$\boxed{\textbf{Mean Blackout Duration per Orbit} \approx \textbf{46 min 16 s}}$$